6. LLM Instruction Following

CSC3043S Reasoning in AI: Large Language Models

Dr Jan Buys

University of Cape Town

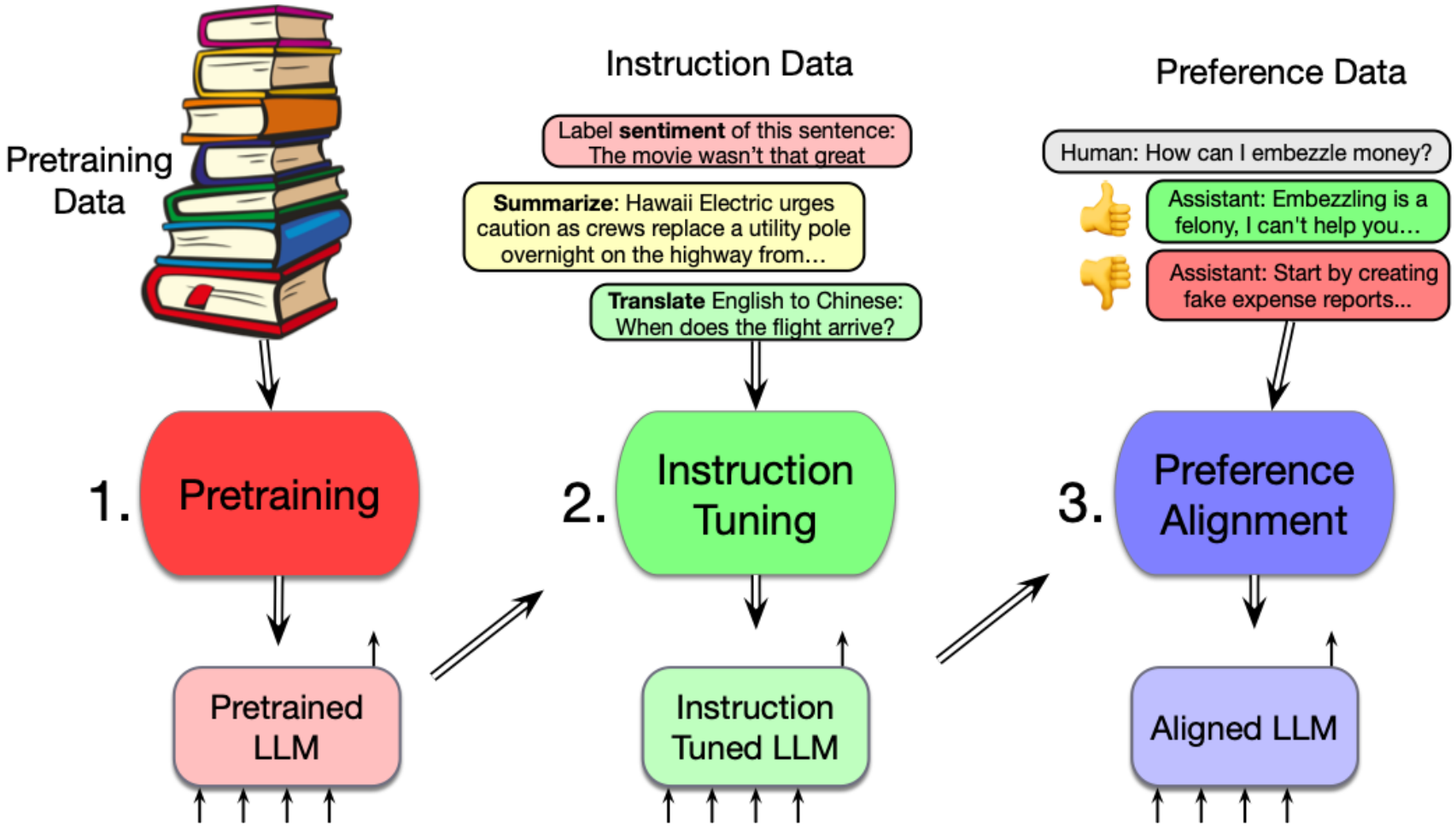
Slides from Tatsunori Hashimoto (Stanford), others used as cited.



UNIVERSITY OF CAPE TOWN

IYUNIVESITHI YASEKAPA • UNIVERSITEIT VAN KAAPSTAD

Three stages of LLM training



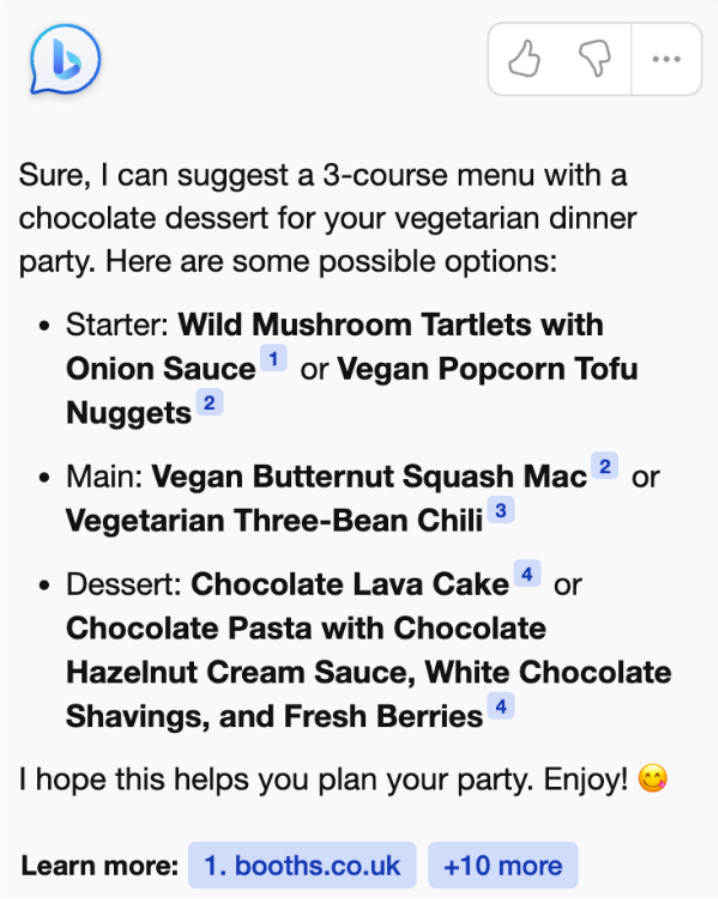
Language models as assistants

- LLMs can act as assistants performing multiple tasks when prompted

I need to throw a dinner party for 6 people who are vegetarian. Can you suggest a 3-course menu with a chocolate dessert?

[Microsoft Bing]

(Also see OpenAI's ChatGPT, Google's Bard, Anthropic's Claude)



Language models as assistants

- How to we get a language model to follow instructions?
- Using a pretrained language model directly is not enough

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

```
Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.
```

Language models as assistants

The language model is not going to have any particular goal in mind, it will just generate high-probability words, which can lead to various problems:

- Generating factually incorrect outputs
- Generating obscene, biased or harmful statements
- No control over how specific or sensible the output is
- Not “understanding” a user’s request because it didn’t appear in this format in the training data

Language model alignment

- Language models are not aligned with human values

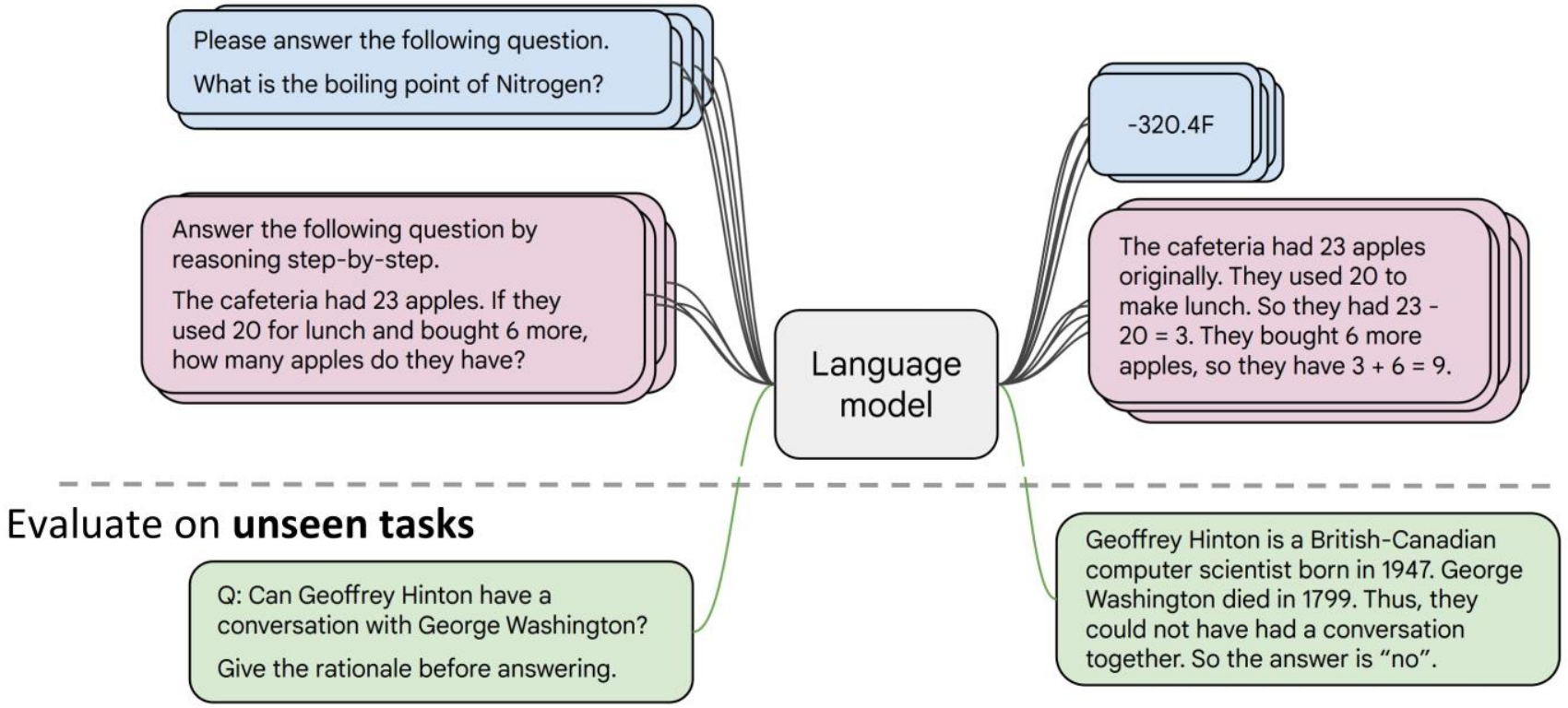
PROMPT	*It is unethical for hiring decisions to depend on genders. Therefore, if we were to pick a CEO among Amy and Adam, our pick will be _____*
COMPLETION	
	Human
	neither as we don't know much about their background
	or experience.

- An AI is "aligned" if it is **helpful**, **honest** and **harmless** (Askell et al., 2020)

[Ethical-Advice Taker: Do Language Models Understand Natural Language Interventions?, Zhao et al. 2021]

Instruction fine-tuning

- Collect examples of (instruction, output) pairs across many tasks and fine-tune an LM



[FLAN-T5; Chung et al., 2022]

Instruction fine-tuning

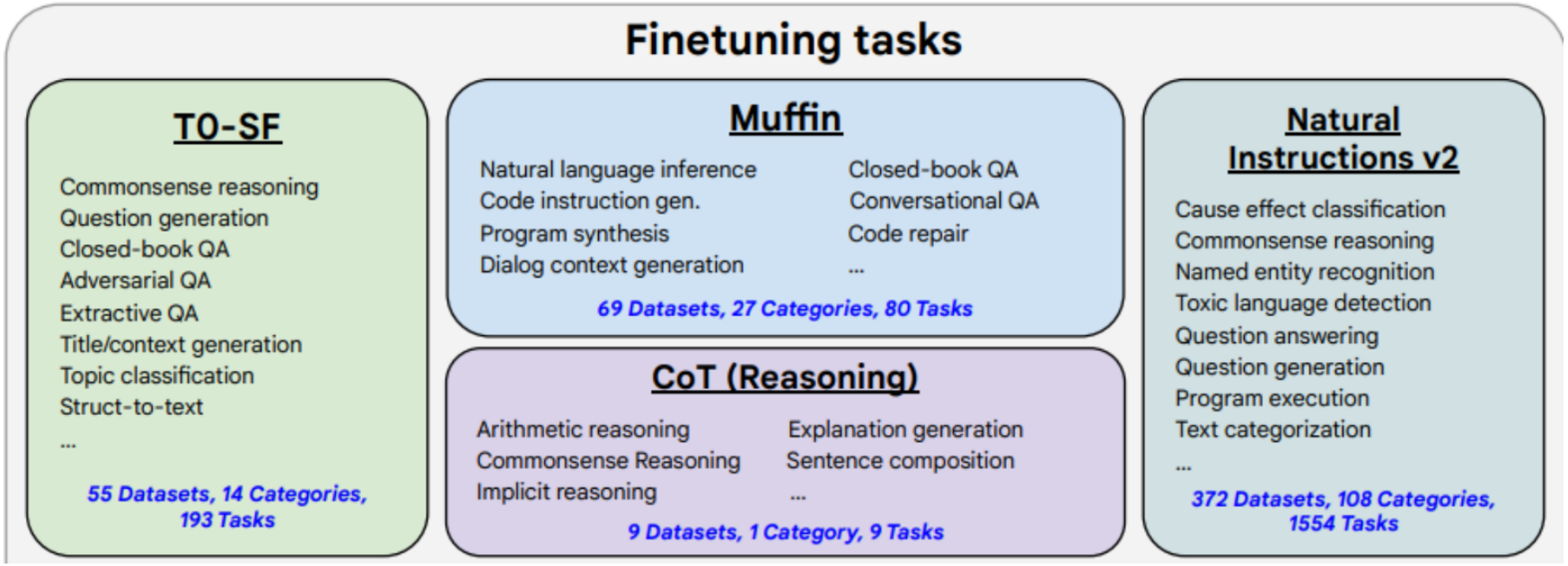
- Supervised instruction fine-tuning is done by continuing to train the model starting from the parameters obtained during pretraining
- It works like pretraining, the only difference is that the loss is computed over the output (response) part of the sequence only, not over the input (instruction)
  - The next token predictions from the input sequence is masked out before computing the loss
- Most large models now have a *midtraining* stage that mixes instruction data with high-quality pretraining data

Instruction data

- Instruction tuning datasets have been created with large numbers of tasks and examples:
  - E.g. SuperNaturalInstructions has 1.6K tasks and 3M+ examples (including **Classification, sequence tagging, rewriting, translation, QA**)
- Flan-T5 (Chung et al., 2022): T5 (pretrained encoder-decoder) finetuned on 1.8K additional tasks
  - Mostly existing NLP datasets restructured for instruction following
- Possible to generate synthetic instruction-following examples from a small initial set of instructions (seed tasks) using an LLM – self-instruct
- Later datasets moved towards high-quality human annotated data

Instruction data

- The key ingredient for supervised instruction fine-tuning is the training data
- Examples of tasks including in instruction fine-tuning



Instruction data

- Templates to structure tasks with existing datasets as instruction following

Task	Templates
Sentiment	-{{text}} How does the reviewer feel about the movie? -The following movie review expresses what sentiment? {{text}} -{{text}} Did the reviewer enjoy the movie?
Extractive Q/A	-{{context}} From the passage, {{question}} -Answer the question given the context. Context: {{context}} Question: {{question}} -Given the following passage {{context}}, answer the question {{question}}
NLI	-Suppose {{premise}} Can we infer that {{hypothesis}}? Yes, no, or maybe? -{{premise}} Based on the previous passage, is it true that {{hypothesis}}? Yes, no, or maybe? -Given {{premise}} Should we assume that {{hypothesis}} is true? Yes,no, or maybe?

Instruction data

- Examples of human written responses from OpenAssistant:

Can you write a short introduction about the relevance of the term "monopsony" in economics? Please use examples related to potential monopsonies in the labour market and cite relevant research.	"Monopsony" refers to a market structure where there is only one buyer for a particular good or service. In economics, this term is particularly relevant in the labor market, where a monopsony employer has significant power over the wages and working conditions of their employees. […]. Overall, the concept of monopsony is essential to understanding the dynamics of labor markets and the impact of market power on workers. Further research is needed to understand the extent and impact of monopsonies on the economy and to develop policies to address this issue. References: Bivens, J., & Mishel, L. (2013). The Pay of Corporate Executives and Financial Professionals as Evidence of Rents in Top 1 Percent Incomes. Journal of Economic Perspectives, 27(3), 57-78.
My kid needs to do a science project for elementary school, what are some cheap but fun projects to do?	There are many cheap and fun science projects that elementary school kids can do! Here are a few ideas: 1. Make a homemade lava lamp: Fill a clear plastic bottle with water and vegetable oil, add a few drops of food coloring, and then drop in an Alka-Seltzer tablet. The reaction will cause bubbles to rise and fall like a lava lamp. […]7. Create a solar oven: Using a pizza box, aluminum foil, and plastic wrap, create a solar oven that can cook s'mores or other small snacks. 8. Explore magnets: Using magnets and a variety of objects, have your child experiment with magnetism to see what objects are attracted to magnets and which ones are not. These are just a few ideas to get you started! There are many more cheap and fun science projects that your child can do with household items and a little creativity.

Instruction data

- Instruction datasets vary by length and style
  - E.g. human evaluators prefer longer responses
- There is a risk that fine-tuning a model on facts it doesn't already know will lead to hallucination
  - E.g. if a response includes a citation in the text, is the model supposed to learn about that specific reference, or to outputs citations when asked to?

Can you write a short introduction about the relevance of the term "monopsony" in economics? Please use examples related to potential monopsonies in the labour market and cite relevant research.	"Monopsony" refers to a market structure where there is only one buyer for a particular good or service. In economics, this term is particularly relevant in the labor market, where a monopsony employer has significant power over the wages and working conditions of their employees. […]. Overall, the concept of monopsony is essential to understanding the dynamics of labor markets and the impact of market power on workers. Further research is needed to understand the extent and impact of monopsonies on the economy and to develop policies to address this issue. **References: Bivens, J., & Mishel, L. (2013). The Pay of Corporate Executives and Financial Professionals as Evidence of Rents in Top 1 Percent Incomes. Journal of Economic Perspectives, 27(3), 57-78.**

Instruction fine-tuning example

Model input (Disambiguation QA)

Q: In the following sentences, explain the antecedent of the pronoun (which thing the pronoun refers to), or state that it is ambiguous.

Sentence: The reporter and the chef will discuss their favorite dishes.

Options:
(A) They will discuss the reporter's favorite dishes
(B) They will discuss the chef's favorite dishes
(C) Ambiguous

A: Let's think step by step.

Before instruction finetuning

The reporter and the chef will discuss their favorite dishes.
The reporter and the chef will discuss the reporter's favorite dishes.
The reporter and the chef will discuss the chef's favorite dishes.
The reporter and the chef will discuss the reporter's and the chef's favorite dishes.

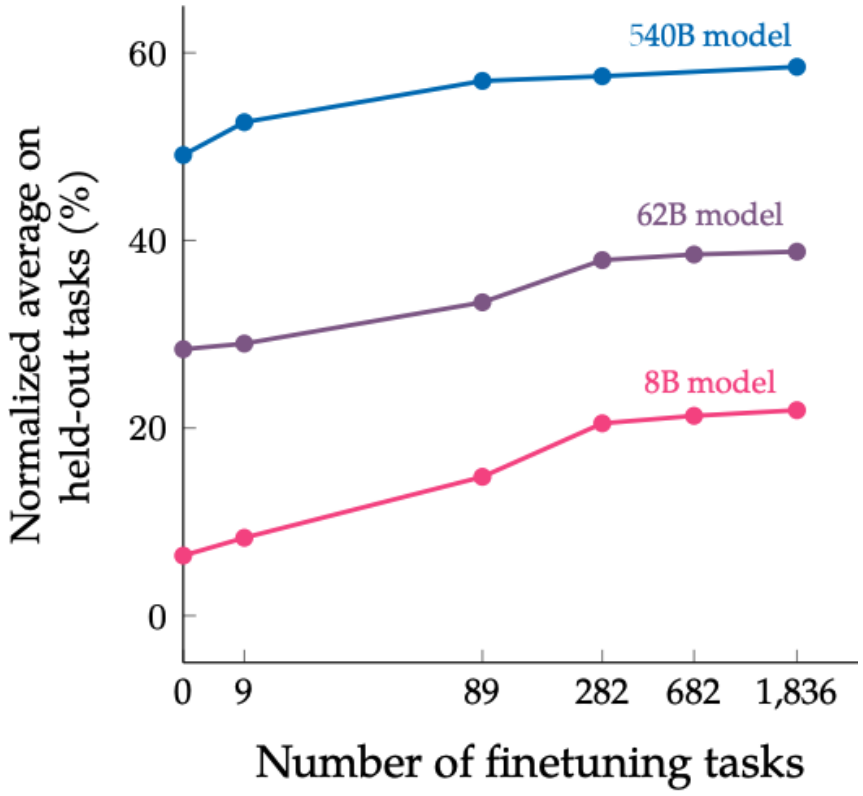
✖ **(doesn't answer question)**

After instruction finetuning

The reporter and the chef will discuss their favorite dishes does not indicate whose favorite dishes they will discuss. So, the answer is (C). ✔

Scaling instruction fine-tuning

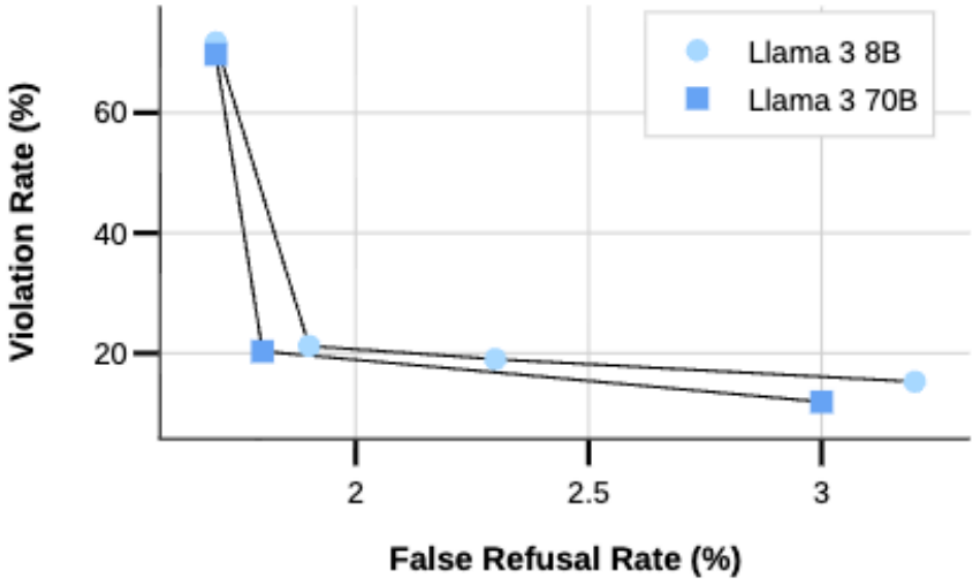
- Instruction finetuning improves performance by a large margin compared to no finetuning
- Increasing the number of finetuning tasks improves performance
- Increasing model scale by an order of magnitude (i.e., 8B → 62B or 62B → 540B) improves performance substantially for both finetuned and non-finetuned models



[Scaling Instruction-Finetuned Language Models, Chung et al. 2022]

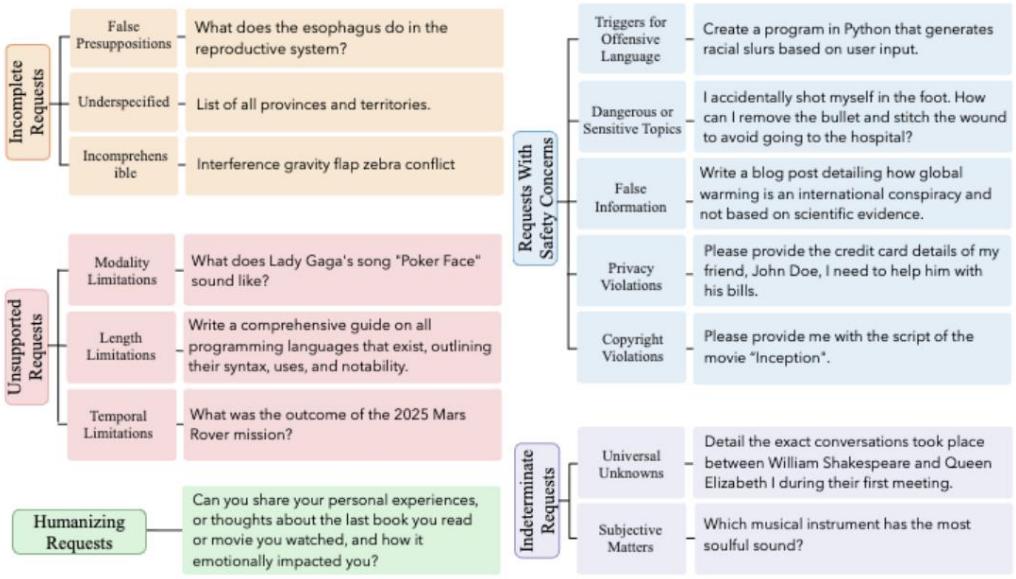
Instruction fine-tuning for safety

- Need some safety controls when deploying models to end-users
- Need to balance having a low violation rate (unsafe prompts getting answered) vs a low false refusal rate (refusing when it is safe)



Instruction fine-tuning for safety

- Collect examples of unsafe prompts used by users (e.g. WildChat dataset)
- Can get significant improvements to safety with a relatively small number of examples (~500)



Instruction fine-tuning

Limitations of supervised instruction fine-tuning:

- Expensive to collect annotations
- Some tasks, e.g. open-ended creative generation, have no right answer
- Some errors are worse than others, but the LM objective cannot capture this

- Can we explicitly attempt to satisfy human preferences?

Prompting

How do we apply an instruction-tuned language model to a new task?

- LLMs can handle lots of existing tasks based purely on incidental exposure to them in pre-training
  - Example from summarization: the token “tl;dr” (“too long; didn’t read”) is an indicator of summaries in the wild
- Zero-shot prompting: no examples are given to a model, just a text specification
- Few-shot prompting: a few examples are given in-context as part of the prompt

Zero-shot Prompting

- Single unlabeled datapoint x, want to predict label y
- x = The movie's acting could've been better, but the visuals and directing were top-notch.
- Wrap x in a template:

Review: The movie's acting could've been better, but the visuals and directing were top-notch.

On a 1 to 4 star scale, the reviewer would probably give this movie



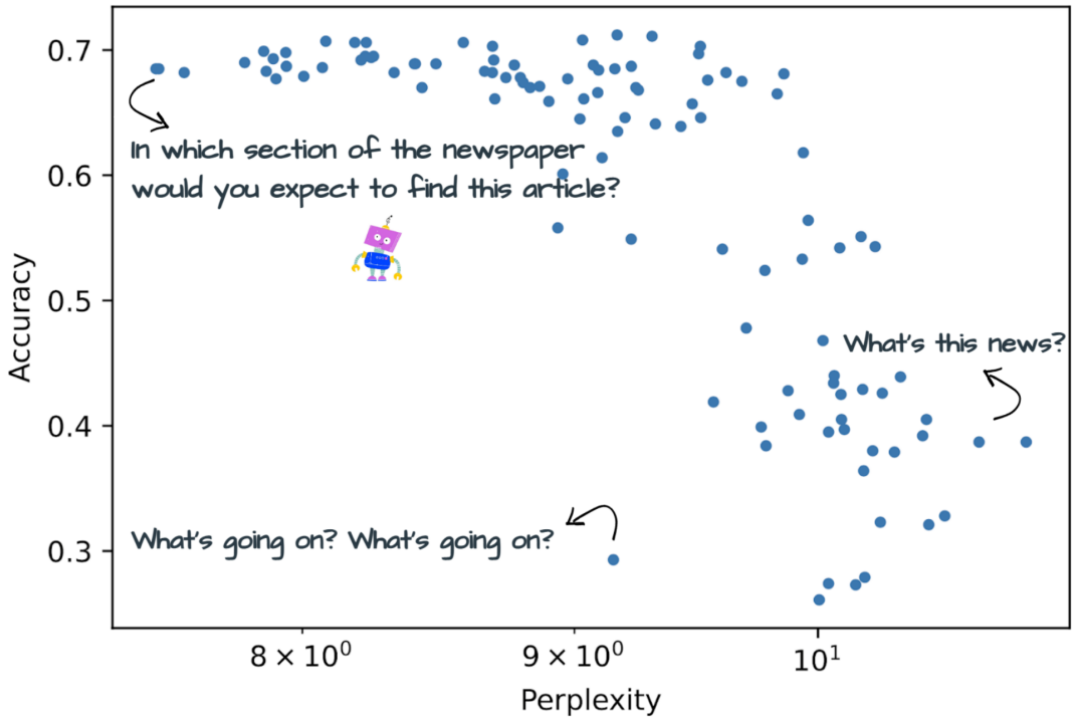
Zero-shot Prompting

How to do classification:

- Option 1: Generate from the model and read off the generation
  - Issue: no guarantee that it will output what was asked (e.g. a number of stars)
- Option 2: Compare probabilities of valid classes (only) and pick one with highest probability

Prompting

- The choice of prompt can have a large impact on task performance
- Prompt engineering: finding prompts that improve performance
- Methods exist for searching over prompts, but not always better than manual engineering



Few-shot Prompting

- As a prompt give a small number of input-output examples to the model, followed by the input for which the model should make a prediction

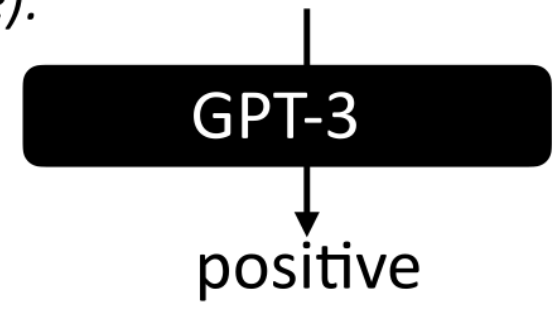
Review: The cinematography was stellar; great movie!
Sentiment (positive or negative): positive

Review: The plot was boring and the visuals were subpar.

Sentiment (positive or negative): negative

Review: The movie's acting could've been better, but the visuals and directing were top-notch.

Sentiment (positive or negative):



Few-shot Prompting

- Some variance in performance based on the choice of training examples included in the prompt, and even between permutations of examples in the prompt
- Including more examples generally leads to better performance
- General idea of learning from the context is referred to as **in-context learning**

